

Transistor BC547 (T2) is used as a pulse amplifier to amplify the signal output from RX1 and, hence, raises the pulse height to CMOS compatible logic -1 (> 3.5V at 5V). This compatible output is then fed to CMOS NAND gate 4011 (IC19). NAND gate N1 helps to get pulses of perfect rectangular-wave shape. Output of gate N1 of IC19 is fed to decoders HT12D (IC20 and IC21). Address lines of the decoders are preset to receive data from encoders IC4 and IC5, respectively.

LEDs (LED5 and LED6) connected at their VT outputs (pin 17) flicker to indicate reception of valid data. Decoding speed (as measured at pin 15) is 200kHz (approximately). Decoded data is then fed to IC22, a DAC (0808). Analogue current output of the DAC (pin 4) is loaded with resistor R26. Voltage developed across it is fed to a digital multimeter, which shows the temperature on mV scale.

A thumbwheel switch (TWS1) is used to change the preset address of the decoders (IC20 and IC21). The switch changes the last three LSB of the address. For example, to get data from the sensor (IC13) connected at IN4 input of the ADC, the number to be set on TWS1 is 4.

Power supplies. The thermometer needs power supplies both for its transmitter and receiver units. Whereas the transmitter unit needs only a +5V regulated power supply built around X1, D1, D2 and IC8 (Fig. 2), the receiver unit needs a dual +5V and -5V supplies built around X2, BR1, IC17 and IC18 (Fig. 3). Both circuits work from 230V AC, 50Hz connected to the primaries of transformers X1 and X2, respectively.

Power supply circuits are self-explanatory. Three-terminal positive voltage regulator IC 7805 is used both in the transmitter and the receiver units to get regulated +5V supplies. On the other hand, a three-terminal negative-voltage regulator IC 7905 is used to get -5V supply required for DAC0808 of the receiver unit.

Construction and testing

An actual-size, single-side PCB pattern for the transmitter unit of the wireless addressable digital thermometer is shown in Fig. 4 and its component layout in Fig. 5. Similarly, an actual-size, single-side PCB pattern for the receiver unit is shown in Fig. 6 and its component layout in Fig. 7.

For connection of TWS1, use a four-wire cable to connect it to the PCB. Fix TWS1 on the front panel of the receiver unit.

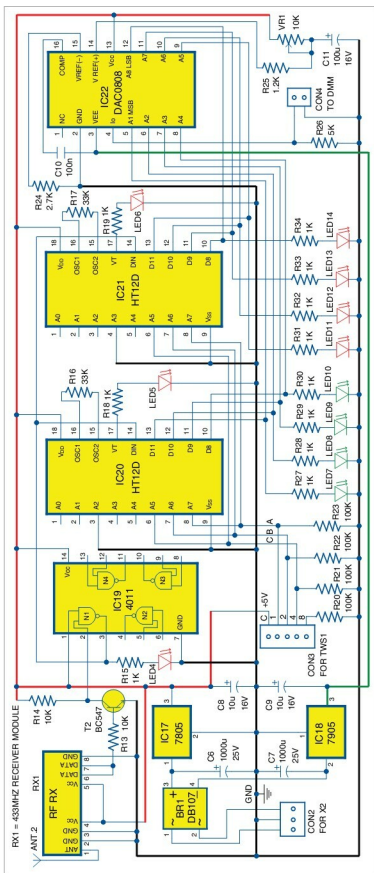


Fig. 3: Circuit diagram of the receiver unit